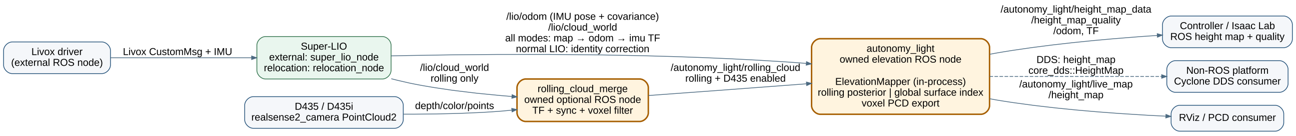
autonomy_light software architecture

Super-LIO integration and elevation mapping · 2026-08-04

> **Design decision.** SLAM and relocalization are Super-LIO responsibilities. This package owns the elevation node and an optional, isolated D435 rolling-cloud merge node. Super-LIO remains the only SLAM backend.



Node responsibilities

Node	Owner	Responsibility
Livox driver	external	Publishes one Livox CustomMsg and IMU stream.
Super-LIO	external	LiDAR–IMU odometry with ESKF pose covariance, Scan Context + GICP + iSAM2 full SLAM, saved-map relocation, and continuous `map → odom → imu` TF.
`rolling_cloud_merge`	this package, optional	Synchronizes one D435/D435i PointCloud2 to a LiDAR cloud, transforms it into `odom|map`, and voxel filters it.
`autonomy_light`	this package	Base pose/TF, rolling or global elevation map, quality output, live PCD and PCD save.

Data and TF contract

Direction	Topic	Meaning
input	`/lio/odom`	Super-LIO IMU pose and 6×6 ESKF covariance; exactly converted to `base_link`.
input	`/lio/cloud_world`	Registered cloud in the current global frame.
optional input	`/camera/.../depth/color/points`	One D435/D435i cloud; used only by the merge node.
intermediate	`/autonomy_light/rolling_cloud`	Time-aligned, global-frame LiDAR+D435 cloud for rolling elevation.

output	`/autonomy_light/height_map_data`	Controller distance grid.
output	`/autonomy_light/height_map_quality`	Per-cell validity, variance, age.
output	`/autonomy_light/live_map`	Sparse voxel PCD for RViz and map saving.
optional output	`height_map`	Direct Cyclone DDS `core_dds::HeightMap` payload for a non-ROS peer.

TF is `map → odom → imu → base_link → base_link_gravity`, with static `imu → base_link → lidar_link`. Super-LIO publishes identity `map → odom` before its first LiDAR update, then owns the correction; this runtime owns the two physical static links. Normal LIO keeps that correction at identity. A saved map runs Super-LIO `relocation_node`. After its NDT→ICP registration succeeds, Super-LIO re-broadcasts `map → odom → imu` at `publish_rate_hz`; the correction is only recalculated at the configured low global-registration rate. Mapping runs the pose graph in Super-LIO. There is no `world` frame or duplicate SLAM backend in this package.

`/lio/odom` is the IMU state. The runtime composes `T_base_imu = T_base_lidar · inverse(T_imu_lidar)` from `target_to_lidar_*` and `imu_from_lidar_*`, then applies `T_global_base = T_global_imu · inverse(T_base_imu)`. Its managed Super-LIO process receives the same `T_imu_lidar` calibration.

Elevation source selection

Configuration	Algorithm	Use
`height_map.source: rolling`	Pose-following local Bayesian ground/upper window with measurement variance and outlier gate.	Default for control and sim-to-real.
`height_map.source: global`	Persistent, PCD-derived sparse surface index.	Repeatable terrain reference from a saved or accumulated map.

Both sources use the same output message. `valid=0` means the distance value is an unknown placeholder, not measured flat ground; Isaac Lab should consume the quality channels.

With `rolling_merge.enabled: true`, the launcher starts the optional merge node. It accepts only a D435 sample within the configured synchronization window, uses TF at that camera timestamp, and passes a LiDAR-only cloud through if TF or a matching camera sample is unavailable.

`base_link_gravity` shares the exact `base_link` origin and removes only roll/pitch from its orientation. It is not translated to an estimated floor, so hanging geometry cannot move the TF. Terrain z is directly base-relative and controller data is `base_z - terrain_z`.

Each rolling cloud is matched to its timestamped odometry covariance. Sensor range and mounting calibration are added to the Jacobian-projected pose covariance; XY uncertainty splats the point across nearby cells. Over-limit uncertainty rejects the cloud. The merge node keeps a `sensor_id` and each

D435 point's transformed origin, so LiDAR and D435 retain independent measurement models and exact cleanup-ray sources.

The rolling map's sparse keys are in the global frame only to avoid coordinate resampling. At every pose the runtime retains exactly the output-sized, yaw-aligned local window and removes cells outside it. Valid terrain therefore persists inside the local window; `age` is a quality diagnostic, not a wall-clock validity timeout.

In rolling mode, per-cloud visibility cleanup casts sensor-to-endpoint rays. A stale surface is cleared only when a ray passes below it after the configured age and aligns with its normal. A distance-ramped exclusion limit removes points above the sensor before fusion, preventing near ceiling and overhang returns from becoming terrain. Static global PCD input is deliberately not ray-cleaned because it has no observation origin or timestamp.

At startup, only the robot footprint receives a flat, high-variance ground prior because its underside is occluded. Its first real observation bypasses the innovation gate and replaces it by the normal Kalman update; the quality variance exposes the prior's low confidence.

`base_link → base_link_gravity` is present from startup as identity. After odometry arrives, only its roll/pitch-removing rotation changes; translation stays zero.

The runtime separately schedules heavy cloud fusion and global-PCL publication, so its output timer refreshes odometry, height-map ROS/DDS messages, and `base_link → base_link_gravity` at `publish_rate_hz`. Static calibration links remain on `/tf_static`. RViz `/autonomy_light/height_map` always contains the entire grid: `intensity=1` means valid, while `intensity=0` is the unknown placeholder.

`./launch.sh --vis` starts a live OpenCV 2D view of only the numeric `HeightMap.data` grid. It keeps the newest sample only and targets 50 Hz, so the image follows robot motion instead of rendering a queued stale frame.

Transport selection

`height_map_output.transport` selects `ros2`, `cyclone_dds`, or `both`. The direct DDS option uses the installed IDL `core_dds::HeightMap { sequence<float> data; }`, configurable domain and topic, and best-effort keep-last delivery. Its flat, row-major data sequence has exactly the same values and unknown convention as ROS `HeightMap.data`; geometry and quality remain ROS-message metadata. `both` is the migration/validation mode.

Removed ownership

- Duplicate local pose graph, Scan Context loop closure, and map reconstruction
- FPFH/GICP initial localization and runtime map correction outside Super-LIO
- Secondary LiDAR merger and custom ROS-domain bridge executors
- Separate raw/refined/ROI map pipelines and duplicated path subscription

These removals make the state path linear: Super-LIO localizes; one node maps elevation.